
Bank Management System

The National Bank Group operates several branches across the country :

- Each branch has its own name, unique branch code, address, contact phone number, and manager who oversees daily operations
- A branch can manage many accounts, but each account must belong to one branch only.
- The bank serves both individuals and businesses.

Each Branch Has One Manager who oversees daily operations.

- Every Manager has a unique identification, along with personal details such as full name, email address, phone number, and HireDate.

Customers usually visit the nearest branch to open and manage their accounts, though they may use services from any branch once registered in the system.

- Every customer has a unique identification number, along with personal details such as full name, date of birth, national ID, email address, Phone number, and home address.
- Some customers are individuals, while others represent businesses or organizations.

Customers can open one or more bank accounts, and sometimes two or more customers share the same account (such as a joint savings account between business partners).

- Each account in the bank is identified by a unique account number. The system also records details like account type (savings, current, business, etc.), opening date, current balance, and the branch where the account was first created
- An account must always belong to one branch, but a branch can manage many accounts

When customers are linked to accounts, the bank also keeps track the relationship between customers and accounts :

- The date of ownership starts, the ownership type (primary holder or co-holder), and whether the account is active or closed.

For transparency and financial tracking, the bank stores all transactions related to each account

- A transaction includes a transaction number, transaction date, amount, transaction type (deposit, withdrawal, transfer, payment, etc.), and a short description or note about the purpose.
- Each transaction must be tied to the specific account it affects.

The system should maintain a complete transaction history so that account statements can be generated at any time

Whenever a new account is created, the system verifies that the branch exists and that the customer(s) linked to the account are already registered. If two customers open a joint account, both should appear in the system as account owners. When a transaction occurs, it automatically adjusts the account's balance based on the transaction type

- Deposits increase balance
- Withdrawals and payments decrease balance
- Confirm the branch exists
- Confirm all customers involved are registered

Managers at each branch are responsible for monitoring all accounts under their branch and ensuring the accuracy of recorded transactions

- In some cases, a single customer may hold accounts across different branches, and the system must maintain accurate and consistent customer data across all branches.